Chapter 5

Adaptation to FTPM, CS and NR

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Preface

This chapter is a preliminary version of the paper in preparation for submission to a peer-reviewed journal: **"Adaptation to frequency-to-place factors in cochlear implant users: longitudinal case studies of vowel and chord perception."**.

Chapters 3 and 4 examined whether individual differences in FTPM, CS, and NR shape the perception of vowels and chords in CI users who have been listening with their device for years.

In the vowel identification task (Chapter 3), we found that both experienced CI groups showed a displacement of their vowel perception relative to NH listeners. However, the displacement was quantitatively small and restricted to a shift of F2 towards low frequencies for the Post-LCIs and a greater reliance on F1 for the Pre-ECIs. None of the frequency-to-place factors studied here explained these small differences. Such absence of relationship could have two meanings: first, it could indicate that experienced users had fully adapted to these factors; second, it could indicate that these factors do not affect perception within the frame of the present task, regardless of adaptation.

In the chord preference task (Chapter 4), interval distortion - corresponding to Analysis 1 and a disruption of the harmonic organisation introduced by the FTPM profile - significantly predicted chord preference, but only through an interaction with chord inversion ($p < .001$ for Post-LCIs; $p = .003$ pooled): distortion sensitivity was detectable only on root-position trials, where the root chord occupied the reference position, and was absent on first-inversion trials, where root-position preference dominated regardless of distortion level. This effect was consistent across Post-LCIs (all five showing negative interaction coefficients) but absent in Pre-ECIs, for whom distortion avoidance was either unconditional (one participant) or absent entirely (one participant). Similarly in Analysis 2 (at the electrode level), FTPM predicted perceptual weighting in the expected direction - less absolute mismatch associated with higher electrode preference for the Post-LCIs. CS was statistically significant for Post-LCIs, with narrower CS associated with higher electrode preference. NR was not significant in either group.

Together, these findings suggest that, for the vowel task, experienced CI users have either adapted to these frequency-to-place factors or that the task is not sufficiently sensitive to detect their perceptual influence (or both). For the chord task, they suggest that even after years of CI use, perception remains constrained by some of these factors - particularly for Post-LCIs.

The study presented in this chapter aims to clarify the adaptation process during the first months following device activation. This work constitutes a preliminary observation of how perception changes with brain adaptation following CI activation.

For the vowel task, the central question is whether the task is genuinely sensitive to these factors: if it is not, we should observe no relationship between the factors and vowel task performance even during the active adaptation period, confirming that the paradigm is simply not suited to tracking them.

For the chord task, we are interested in whether sensitivity to distortion changes over the months following activation. If it does not change, it would suggest that the perceptual constraints imposed by these factors are persistent, and that only a clinical remapping intervention could alleviate them. If sensitivity decreases over time, it would

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suggest that some capacity for adaptation exists, with the auditory system progressively accommodating the spectral distortions imposed by the factors studied.

This chapter addresses these questions through a single longitudinal case study of a newly implanted adult Post-LCI user, tested repeatedly during the first weeks and months post-activation using the same vowel identification and chord preference tasks as Chapters 3 and 4. FTPM was estimated from post-operative X-ray imaging, and CS and NR were estimated from PECAP recordings, providing electrode-level frequency-to-place factor data for all three measures.

Only one case study is presented here; three additional longitudinal participants are currently being tested and will be incorporated into the final version of this work. The goal is therefore not to draw population-level conclusions from a single individual, but to characterise the temporal dynamics of adaptation and to generate hypotheses about the mechanisms that determine whether, and how quickly, CI users adapt to individual frequency-to-place factors.

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Adaptation to frequency-to-place factors in cochlear implant users during the first months of cochlear implant use: longitudinal case study of vowel and chord perception

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Abstract

Understanding how CI users adapt to their device in real time requires longitudinal paradigms sensitive enough to detect perceptual change at the electrode level. This chapter tracks the perceptual representations of a single newly implanted Post-LCI across four sessions spanning Days 6 to 134 post-activation, using the vowel identification and confidence task from Chapter 3 and the chord preference paradigm from Chapter 4. In the vowel task, F1 and F2 confidence representations co-varied across sessions for most vowels, with F2 drifting monotonically away from the NH reference towards more basal cochlear positions - opposite to the convergence expected from adaptation. One vowel, [ɜː], showed independent F1 movement consistent with genuine electrode-to-category learning. The F1/F2 reliance ratio showed no temporal trend, with the Day 6 spectral weighting profile already characteristic of a Post-CLIs, suggesting that spectral reweighting is not a mechanism of early adaptation. Electrode-level factor analyses revealed a coherent temporal trend at F2, whereas no such temporal trend was seen for F1. At F2, electrodes with more absolute FTPM, narrower CS and smaller NR attracted more confidence in the early time of the adaptation, and trend reversed gradually overtime. In the chord task, FTPM distortion sensitivity was present at Days 6 and 20 but had disappeared by Day 41, giving way to electrode-level FTPM preference by Day 134. NR facilitated preference formation in the early sessions, whilst FTPM emerged as the dominant organisational factor in the extended phase. Across both tasks, the period between Days 20 and 41 marked a qualitative transition in the representational basis of perception, from acoustic template matching to electrode-level organisation. These findings characterise a two-phase model of early CI adaptation and demonstrate that the chord paradigm is more sensitive than the vowel task for tracking this process in real time.

5.1 Introduction

The preceding chapters characterised how experienced CI users represent vowels and chords at the individual electrode level, and identified which frequency-to-place factors

shape that representation after years of device use. In Chapter 3, both Post-LCIs and Pre-ECIs showed only small displacements in their electrode confidence distributions from the NH reference. Critically, the delta analysis revealed that this displacement reflects a global reorganisation of spectral weighting rather than electrode-by-electrode degradation: at matched frequency positions, none of the frequency-to-place factors predicted the local deviation from NH, and this null held within both groups independently. The distinction is important: a spectral weighting account implies that the entire confidence profile has been uniformly shifted or rescaled relative to NH, whereas an electrode-by-electrode account would predict that specific electrodes - those with the greatest mismatch or poorest neural interface - would show localised deficits. The absence of electrode-specific effects in Chapter 3 is consistent with the former but does not rule out the latter during the early adaptation period. In Chapter 4, FTPM-induced interval distortion affected chord quality judgements in Post-LCIs, and FTPM predicted perceptual preference weights at the electrode level; CS was marginally significant for Post-LCIs; NR was not significant for affecting quality judgements in either group.

Together, these findings might characterise the perceptual endpoint of the adaptation process - the state at which experienced CI users distribute confidence across the electrode array and weight chords after years of exposure to a spectrally distorted input. However, two important ambiguities remain that cross-sectional designs cannot resolve. For the vowel task, the null effect of frequency-to-place factors on the delta confidence profile could reflect either genuine completion of adaptation - meaning electrode-level factors no longer drive any local deficit - or simply that the paradigm is insensitive to residual effects. For the chord task, it is unclear whether the residual influence of FTPM (and marginally CS) on preference reflects a fundamental limit on adaptation, or a transient state that would resolve with further experience. Disentangling these possibilities requires testing users before adaptation has taken place, at repeated intervals across the early post-activation period, before adaptation has finalised.

Indeed, this early post-activation period is the most dynamic phase of CI use. Most perceptual improvement occurs within the first three to six months following device activation, with gains progressively flattening thereafter (Fu and Galvin, 2007a; Holden et al., 2013a). At the level of electrode discrimination, Mathew et al. (2018) showed that both electrophysiological and behavioural measures of channel selectivity improve over the first six months, with the rate of improvement correlating with speech perception gains - suggesting that electrode-level perceptual changes track the broader adaptation trajectory. Despite its clinical importance, the mechanisms at the electrode level driving this improvement remain poorly understood: standard clinical management focuses on adjusting stimulation levels and collecting subjective feedback, with electrode-level factors not yet systematically integrated into fitting practice (Holden et al., 2013a).

Of the three frequency-to-place factors examined in this thesis, FTPM is the best characterised in the adaptation context. The foundational study by Fu and Galvin (2003) demonstrated that CI users can partially adapt to experimentally imposed frequency shifts over three months, but performance with the shifted processor never recovered to clinical-map levels, and subsequent computational modelling confirmed that none of the participants achieved complete adaptation (Svirsky et al., 2005). The general consensus is that moderate FTPM can be largely compensated by auditory plasticity when daily use is sufficient, but severe mismatch may substantially prolong the adaptation time course and may never be fully resolved over the first six to twelve months (Fu and Galvin,

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2007a; DeFreese et al., 2025). Critically, however, all existing FTPM adaptation studies reduce mismatch to a single subject-level value - a mean across electrodes - making it impossible to determine which electrodes are driving or limiting the adaptation process at the individual level.

For CS and NR, longitudinal perceptual data are essentially absent. CS is largely fixed following implantation as a property of the electrode-neural interface, and the implicit assumption in the literature is therefore that its perceptual consequences are static throughout adaptation (Brown et al., 2000). This assumption has never been directly tested. NR is also largely stable after the initial post-surgical period (Brown et al., 2000), but its perceptual consequences may not be equally static: unlike CS, which acts as a fixed spectral resolution constraint, inter-electrode variation in NR requires the central auditory system to learn to reweight signals of unequal neural fidelity across the array - an adaptive process whose direction and magnitude may themselves evolve over time. In particular, electrodes with higher NR produce sharper and more consistent signals that may initially be perceived as unfamiliar relative to the internal representation established during the period of deafness, such that the perceptual advantage of high-NR electrodes may only emerge after central recalibration - predicting a non-monotonic NR-perception trajectory that would be invisible to cross-sectional designs.

This chapter addresses three questions that the cross-sectional designs of Chapters 3 and 4 could not answer. The following metrics, introduced in Chapters 3 and 4, are used to track the adaptation trajectory across sessions: the *Wasserstein distance* (overall magnitude of displacement from NH), the *formant centroid shift* (direction of displacement along the frequency axis), the *F1/F2 reliance ratio* (balance of confidence between formant regions), and the electrode-level *delta bell* (signed deviation from the NH reference at each electrode position) for the vowel task; and the *frequency importance function* (electrode-level preference weight) and *interval distortion* (FTPM-induced harmonic disruption) for the chord task.

Question 1: What does the adaptation trajectory look like for the vowel task? As a newly implanted CI user gains experience with their device, does their electrode confidence distribution converge towards the NH reference, towards the experienced Post-LCI average, or along another trajectory? The Wasserstein distance, formant centroid shift, and F1/F2 reliance ratio provide a framework for tracking this representational change across sessions in a way that is directly comparable to the experienced-user baseline established in Chapter 3.

Question 2: Do frequency-to-place factors predict the electrode-level adaptation trajectory for the vowel task? Beyond the group-level trajectory, do individual electrodes characterised by greater absolute FTPM, broader CS, or poorer NR show systematically different patterns of confidence change over time? This question extends the delta confidence analysis of Chapter 3 - which revealed a null effect in experienced users, interpreted as evidence of global reorganisation - to ask whether that null is present from the very first session, or whether factor-driven local deficits are visible during the early post-activation window before global reorganisation has consolidated.

Question 3: Do frequency-to-place factors predict the chord preference adaptation trajectory? Chapter 4 established that FTPM-induced interval distortion affects chord quality judgements in experienced Post-LCIs, and that FTPM magnitude predicts electrode-level perceptual preference weights. Whether these relationships are

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present from the beginning of CI use, or whether they emerge progressively as adaptation consolidates, is unknown. Tracking the distortion-preference relationship and the FIF profiles across sessions in a newly activated user provides a longitudinal complement to the cross-sectional findings of Chapter 4, and may reveal whether early targeted remapping - designed to reduce interval distortion at specific electrode pairs - would be clinically beneficial before chord preferences have consolidated around a suboptimal map.

This chapter presents a longitudinal case study of a newly implanted post-lingually deaf adult CI user, assessed across four sessions at Days 6, 20, 41, and 134 post-activation using the same vowel identification and chord preference tasks developed in Chapters 3 and 4. The design is explicitly exploratory: with a single participant, the analyses are descriptive rather than inferential, and the purpose is to generate theoretically grounded hypotheses for a larger prospective longitudinal study rather than to draw confirmatory conclusions.

5.2 Methods

This chapter presents longitudinal data from a single newly implanted CI user (Case Study 1) who completed the vowel identification task and the chord preference task at four time points during the first months following device activation. The experimental tasks, stimuli, response formats, and estimation of individual factors were identical to those described in Chapters 3 and 4; only aspects specific to the longitudinal protocol are described here.

5.2.1 Participant

Case Study 1 was a post-lingually deafened adult recruited from the Emmeline Centre for Hearing Implants at Addenbrooke's Hospital (Cambridge, UK) prior to CI activation, under the same ethics protocol as the cross-sectional cohort (Section 3.2.1).

Table 5.1: **Clinical details for Case Study 1.**

Sex	Female
Age at activation	26 years
Age at onset of deafness	11 years
Duration of deafness	15 years
Aetiology	Unknown
Device manufacturer and model	Cochlear
Number of active electrodes	22
Implanted ear	Left
Contralateral ear status	Hearing Aid

5.2.2 Longitudinal Design

Case Study 1 was tested at four sessions following device activation, scheduled so as to capture early adaptation (first six weeks) and later consolidation (beyond two months),

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as summarised in Table 5.2.

Table 5.2: **Longitudinal testing schedule for Case Study 1.** Days post-activation, approximate period and tasks completed.

Session	Days post-activation	Approx. period	Tasks
S1	6	Week 1	Vowel, Chord
S2	20	Week 3	Vowel, Chord
S3	41	Week 6	Vowel, Chord
S4	134	Month 4.5	Vowel, Chord

5.2.3 Tasks

The vowel identification task was identical to that described in Chapter 3 (Sections 3.2.2–3.2.3): at each session, Case Study 1 heard synthetic vowel stimuli whose F1 and F2 formant frequencies were set to the centre frequencies of their allocated electrodes, identified the perceived vowel from five response options, and rated their confidence on a five-point Likert scale. The same F1/F2 electrode combinations were presented at every session, enabling within-electrode tracking of confidence change over time. The chord preference task was identical to that described in Chapter 4 (Sections 4.2.2–4.2.3), with the same electrode pairs tested at every session. Using identical stimuli and procedures across sessions was essential to ensure that any observed change in performance could be attributed to perceptual adaptation rather than to differences in task demands or stimulus content.

5.2.4 Estimation of Frequency-to-Place Factors

The three electrode-level factors - frequency-to-place mismatch (FTPM), current spread (CS), and neural responsiveness (NR) - were estimated following the same procedures as Chapter 3 (Section 3.2.4). FTPM was computed from a single post-operative X-ray and treated as constant across sessions, on the assumption that electrode position does not change following implantation. CS and NR were estimated from a single PECAP recording obtained at Session 1 (6 days post-activation) and likewise treated as constant across sessions, consistent with the established stability of ECAP-derived measures after the initial post-surgical period (Brown et al., 2000). Both CS and NR are per-electrode values derived from the PECAP decomposition (see Section 3.2.4), not averages across the array; each electrode therefore has its own CS and NR estimate, enabling the electrode-level analyses reported below. Any apparent change in the factor-perception relationship across sessions therefore reflects central adaptation to a stable peripheral input rather than a change in the peripheral measure itself. A representation of these factors for Case Study 1 is provided in Figure 5.1.

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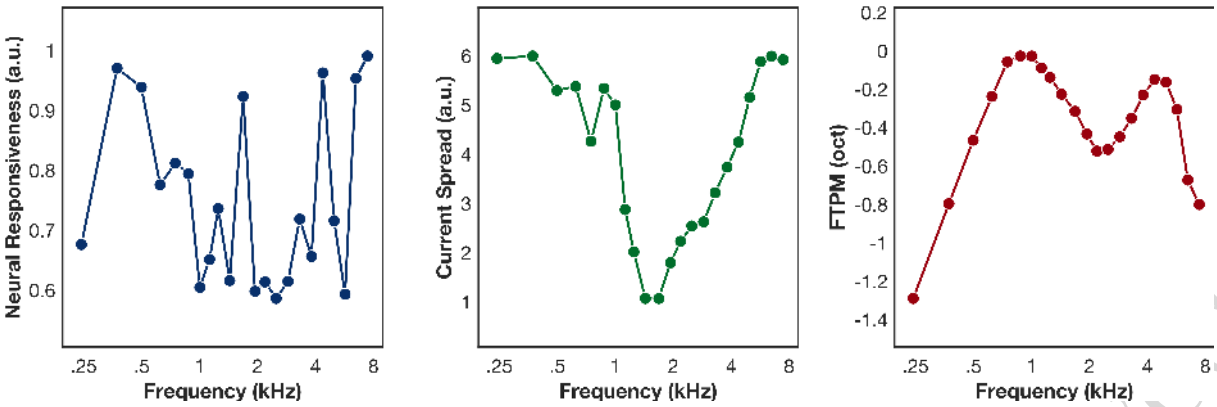


Figure 5.1: **Electrode-level factor profiles for Case Study 1.** Each panel shows the raw value of one factor as a function of electrode centre frequency (kHz, log scale). Each point represents one electrode. Left: neural responsiveness (NR; higher values indicate healthier neural populations). Centre: current spread (CS; higher values indicate broader stimulation and poorer selectivity). Right: frequency-to-place mismatch (FTPM, in octaves; more negative values indicate greater basal shift).

5.2.5 Data Analysis

Exclusion of [iː] responses

As in Chapter 3 (Section 3.2.5), responses for [iː] were excluded from all vowel analyses. All results therefore concern four vowel categories: [ɑː], [ɜː], [ɔː], and [uː].

Electrode Confidence Distribution

The electrode confidence distribution - the *bell* - was computed at each session following the identical procedure described in Section 3.2.5: confidence-weighted kernel density estimation in log-frequency space, with bandwidth $h = 0.10$ $\log_{10}$-Hz and a grid of 300 points spanning 200–9000 Hz. The same NH reference bell computed from the cross-sectional cohort of Chapter 3 was used as the reference throughout, held constant across all four sessions.

5.2.6 Statistical Analysis

All statistical analyses were conducted in MATLAB R2025a. Given the single-participant design, trial-level or grid-point level pooling across sessions was used to provide adequate degrees of freedom within the mixed-effects framework. It should be noted that electrode-level observations within a session are not independent, as electrodes within the same array share a common peripheral and central processing context; the random intercept for electrode in the LMMs accounts for stable between-electrode differences but does not model within-array spatial autocorrelation. Similarly, Spearman correlations reported below are computed on electrode-level observations that are not strictly independent; these are presented as descriptive summaries rather than formal inferential tests. No blanket correction for multiple comparisons was applied; a convergence-of-evidence approach was

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adopted instead, with effects consistent across formant dimension, vowel, and task treated as robust and isolated significant results flagged cautiously. This approach is appropriate for an exploratory single-case design where the goal is hypothesis generation rather than confirmatory inference. All analyses were conducted at $\alpha = .05$. Throughout, results are interpreted against two reference benchmarks: the NH group (representing the perceptual target) and the Post-LCIs from Chapter 3 (representing the asymptotic endpoint of adaptation).

Question 1: What Does the Adaptation Trajectory Look Like for the Vowel Task?

The same three complementary analyses applied to group differences in Chapter 3 were applied here longitudinally, with session replacing group as the primary variable of interest.

Wasserstein Distance Trajectory. The Wasserstein distance from the NH reference bell was computed for each session, vowel, and formant dimension following Equation 3.4, using the same leave-one-out NH baseline. Distances were plotted as a function of days post-activation, separately per vowel and formant, and compared to the NH baseline and the Post-LCI distribution from Chapter 3. A Spearman rank correlation between Wasserstein distance and days post-activation was computed per vowel and formant as a descriptive summary of the trajectory direction.

Formant Centroid Shift Trajectory. The formant centroid shift - $\log_2(\hat{f}_{\text{NH}}/\hat{f}_{\text{session}})$ in octaves, positive values indicating a more apical representation than NH - was computed per session, vowel, and formant dimension following Equations 3.5–3.6. Trajectories were plotted and a Spearman correlation with days post-activation was computed per vowel and formant.

F1/F2 reliance ratio trajectory. The reliance ratio $R = \text{peak}(\text{bell}_{F2})/(\text{peak}(\text{bell}_{F1}) + \text{peak}(\text{bell}_{F2}))$ was computed per session and per vowel following Equation 3.7. The trajectory was plotted per vowel and on average, and compared to the NH reference and Post-LCI values from Chapter 3.

Question 2: Do Frequency-to-Place Factors Predict the Electrode-Level Adaptation Trajectory for the Vowel Task?

Delta Confidence Analysis. The delta confidence measure $\Delta_{ejv} = \text{bell}_{\text{session}}(f_e, v) - \text{bell}_{\text{NH}}(f_e, v)$ was computed per session, electrode, and vowel following Equation 3.3, quantifying how far above or below the NH reference each electrode's confidence lies at each session. The same univariate LMM structure was applied with Δ_{ejv} as the outcome, testing per session whether FTPM, CS, NR, or $\log_2(\text{CF}_{\text{place}})$ predicted the direction and magnitude of the local deviation from NH. A significant negative coefficient at early sessions that diminishes or disappears at later sessions would indicate that factor-driven local deficits are present during the adaptation window and resolve over time - the key prediction motivating this analysis. This per-session delta analysis is the longitudinal

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complement to the group-level delta null result reported in Chapter 3: where the Post-LCIs showed no factor-driven local deficits on average, the present analysis asks whether Case Study 1 shows such deficits early in adaptation before they resolve.

Question 3: Do Frequency-to-Place Factors Predict the Chord Preference Adaptation Trajectory?

Analysis 1 - Interval Distortion and Chord Preference. For each trial, interval distortion was computed following Equation 4.9 from Chapter 4, defined as the summed absolute difference between the cochlear-place interval and the allocated-frequency interval across the two chord electrodes. All trials were recoded relative to the root-position chord following the same sign-flip convention used in Chapter 4: $\Delta d_{\text{root}} = d_{\text{root}} - d_{\text{inv}}$ (negative when root is less distorted) and preference grade positive when root chord preferred.

The models mirror those of Chapter 4, applied per session rather than pooled across participants, all fitted as LMMs with a random intercept for electrode pair. M-Base (*chord type only*) establishes root-position preference at each session. M-Grad (*FTPM gradient only*) tests whether the inter-electrode FTPM gradient (the absolute difference in FTPM between the two chord electrodes) predicts preference strength. M-Add (*distortion + chord type*) controls for root-position preference additively. M-Int (*distortion × chord type*) tests whether distortion avoidance is conditional on chord type:

$$\text{Grade}_{\text{root}} \sim \Delta d_{\text{root}} \times \text{ChordType} + (1 \,|\, \text{PairCat}) \tag{5.1}$$

Longitudinal change was assessed by pooling across sessions with session as a continuous covariate:

$$\text{Grade}_{\text{root}} \sim \Delta d_{\text{root}} \times \text{ChordType} \times \text{Session} + (1 \,|\, \text{PairCat}) \tag{5.2}$$

Three time intervals were analysed: Early (Sessions 1-2, Days 6-20), Extended (Sessions 3-4, Days 41-134), and Full (Sessions 1-4).

Analysis 2 - Electrode Factor Prediction of Chord Preference. This analysis examined how electrode-level factors (FTPM, CS, NR) modulated the longitudinal trajectory of chord preference at the electrode level ($n = 88$: 22 electrodes × 4 sessions). The outcome variable was the electrode-level Bradley-Terry preference score, estimated from the full set of pairwise comparisons involving each electrode using `PairedCompCalc` (Leijon et al., 2019). Each predictor was z-scored prior to model fitting.

Two interaction models were fitted per predictor per time interval - a linear model testing monotonic change over time and a non-linear model allowing session-specific interactions relative to Session 1 (Day 6) as reference:

$$\text{Preference} \sim \text{Time} \times \text{Factor}_{\text{sc}} + (1 \,|\, \text{ElectrodeCat}) \tag{5.3}$$

$$\text{Preference} \sim \text{SessionFactor} \times \text{Factor}_{\text{sc}} + (1 \,|\, \text{ElectrodeCat}) \tag{5.4}$$

where $\text{Factor}_{\text{sc}}$ is the z-scored predictor and `SessionFactor` is a categorical session variable. Model panels display only the interaction component, with all predictor terciles anchored at zero at Session 1, such that divergence between lines reflects purely the time × predictor interaction. The linear model was preferred when its AIC was lower than

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the non-linear model; otherwise the non-linear model was reported. Three epoch subsets were analysed: Early (Sessions 1-3, Days 6-41), Extended (Sessions 3-4, Days 41-134), and Full (Sessions 1-4).

5.3 Results

Results are presented across four testing sessions (S1: Day 6, S2: Day 20, S3: Day 41, S4: Day 134 post-activation), organised around the two tasks administered at each session. Throughout, Case Study 1's trajectories are benchmarked against the NH reference and the Post-LCI group from Chapters 3 and 4.

5.3.1 Vowel Identification Task

Descriptive Overview of Responses

Figure 5.2 shows the perceptual maps for each vowel across the four sessions, alongside the NH reference (right column). Each perceptual map displays the distribution of responses across the F1×F2 electrode grid, with colour intensity reflecting the Bayesian confidence at each grid point. Figure 5.3 shows the corresponding electrode confidence bells, plotting the confidence-weighted kernel density estimate across the frequency axis for each vowel and session. The NH and Post-LCI reference bells are also reported.

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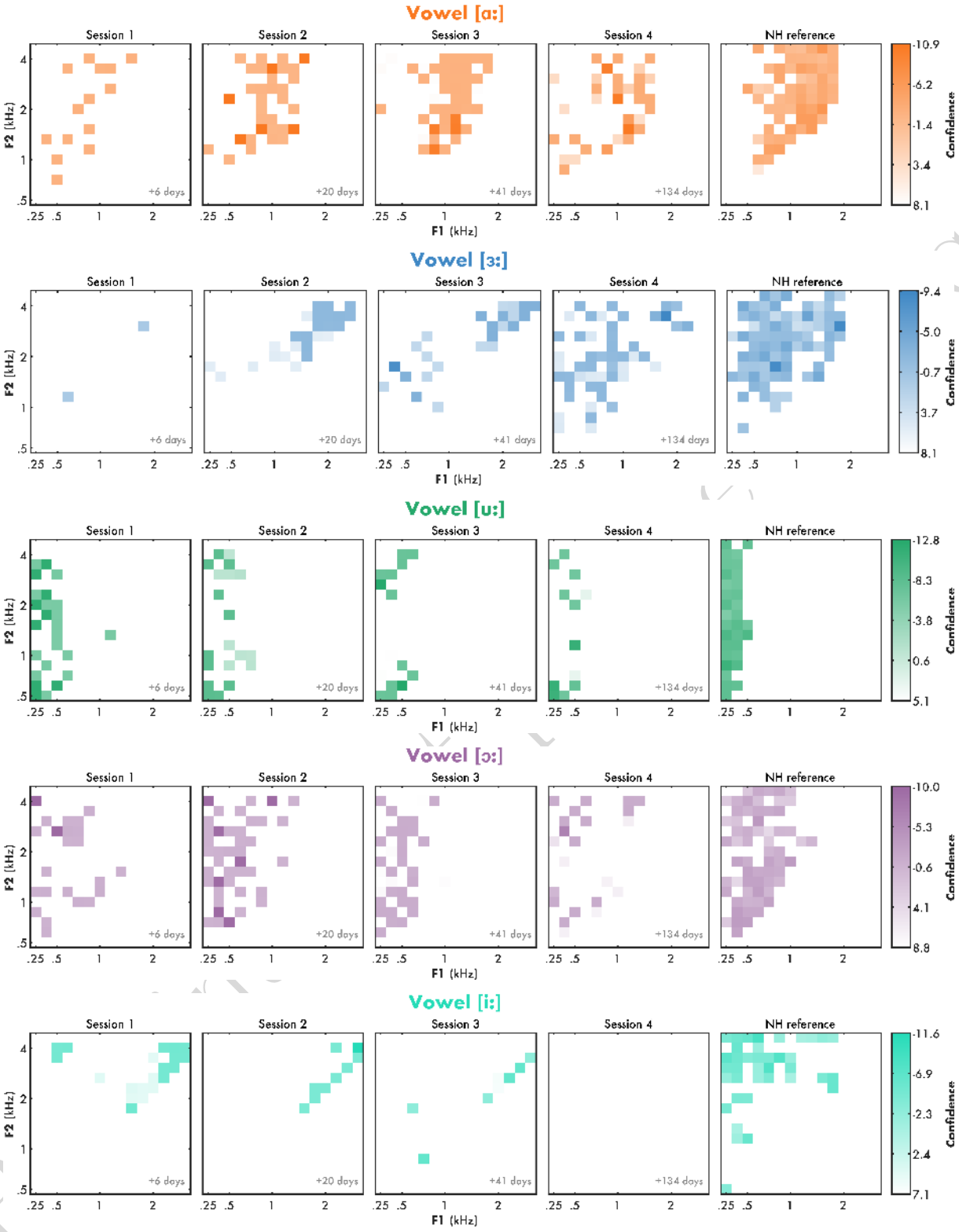


Figure 5.2: **Perceptual maps per vowel across sessions (Case Study 1), alongside the NH reference maps.** Each row shows one vowel (including [iː]) across the four sessions (columns: Day 6, Day 20, Day 41, Day 134). The last column corresponds to the NH group. Colour intensity reflects the confidence of responses at each F1×F2 electrode grid point.

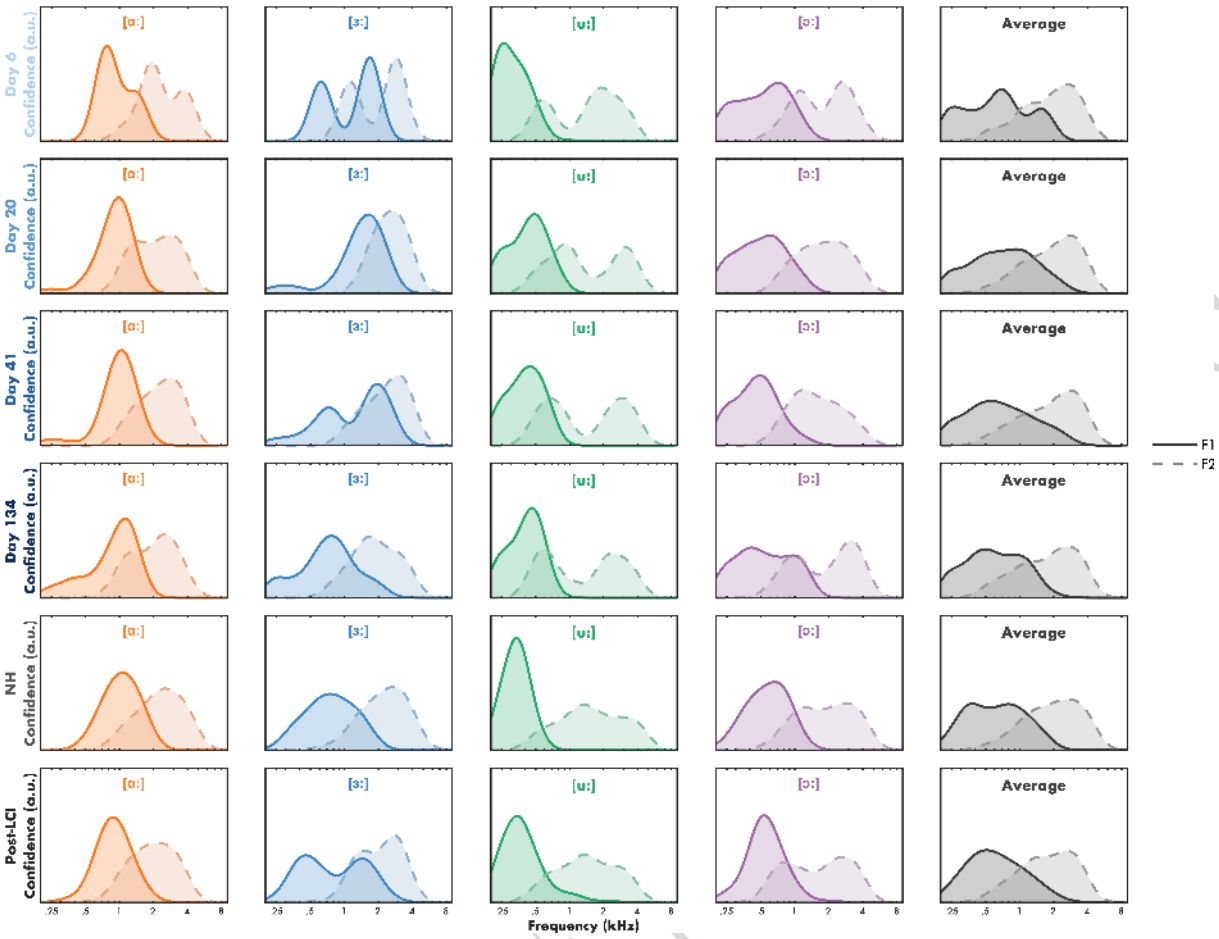


Figure 5.3: **Electrode confidence bells per vowel and session (Case Study 1).** Each panel shows the confidence-weighted kernel density estimate across the frequency axis for one vowel (rows) at each session (coloured lines, light = Day 6 to dark = Day 134). The NH and Post-LCI reference bells are reported in the last two rows.

At S1 (Day 6), responses were broadly distributed across the grid for all vowels, with no clear concentration around the target position. By S4 (Day 134), the response distributions had sharpened for most vowels, with a more clearly defined mode, though the degree of sharpening was vowel-specific. [ɜː] showed the most dramatic change, with a highly dispersed distribution at S1–S2 that consolidated substantially by S4. [ɔː] and [uː] remained relatively diffuse throughout.

Question 1: What Does the Adaptation Trajectory Look Like at the Representational Level for the Vowel Task?

Wasserstein Distance from the NH Reference. Figure 5.4 shows the Wasserstein distance trajectory per vowel and formant, and Table 5.3 reports numerical values.

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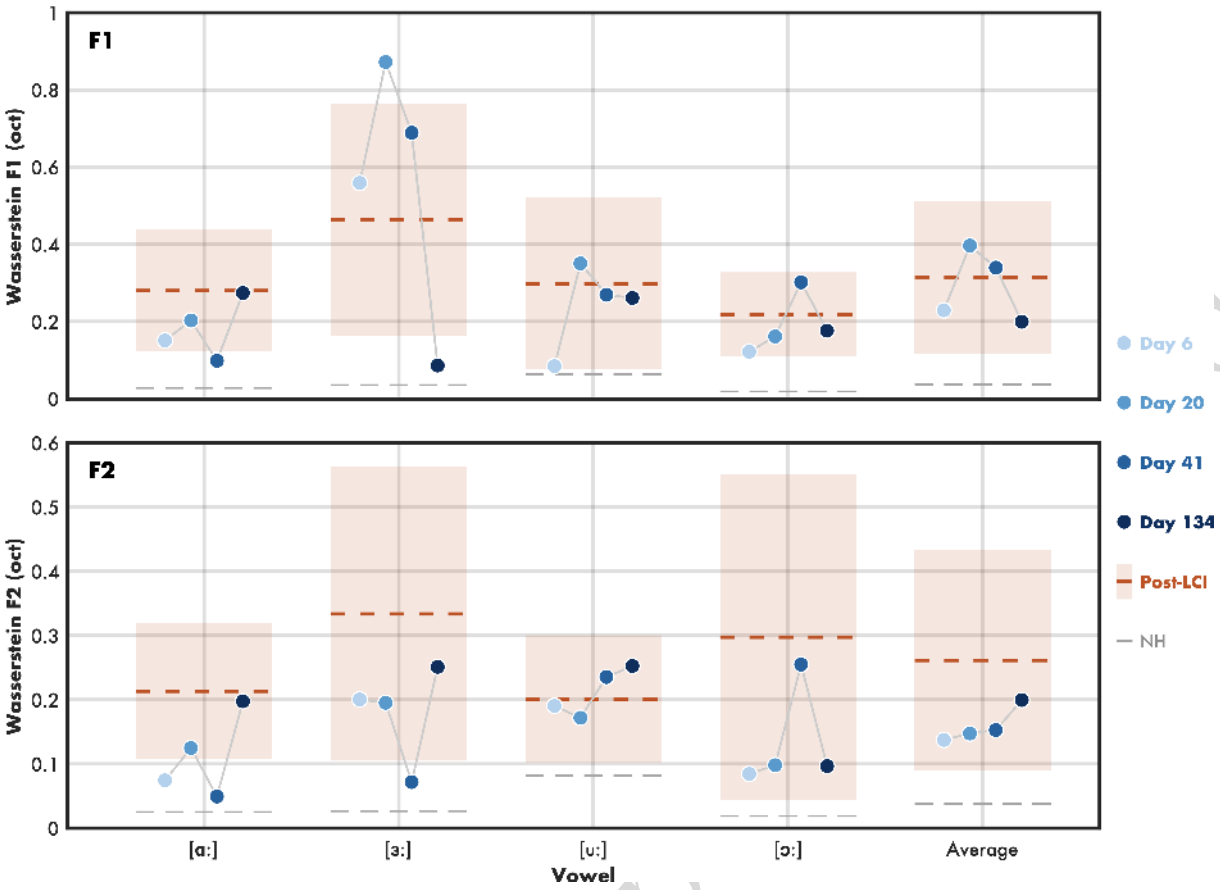


Figure 5.4: **Wasserstein distance from the NH reference bell across sessions (Case Study 1).** F1 (top) and F2 (bottom) distances per vowel and session. The grey dashed line shows the NH baseline; the red dashed line with shaded band shows the Post-LCI mean ± SD. Dots are shaded from light (Day 6) to dark (Day 134).

Table 5.3: **Wasserstein distances from the NH reference bell (oct), Case Study 1.** Values shown per formant, vowel, and session.

Formant	Vowel	S1 (d6)	S2 (d20)	S3 (d41)	S4 (d134)
F1	[ɑː]	0.151	0.203	0.099	0.274
	[ɜː]	0.559	0.872	0.689	0.086
	[uː]	0.085	0.351	0.270	0.261
	[ɔː]	0.122	0.162	0.302	0.177
	Average	0.229	0.397	0.340	0.200
F2	[ɑː]	0.075	0.125	0.049	0.198
	[ɜː]	0.201	0.195	0.072	0.251
	[uː]	0.190	0.172	0.235	0.253
	[ɔː]	0.085	0.098	0.255	0.097
	Average	0.138	0.148	0.153	0.200

F1 Wasserstein distances were dominated by [ɜː], which showed the largest displacement from NH across all sessions, peaking at 0.872 oct at Day 20 before converging sharply to 0.086 oct at Day 134 - the lowest F1 value recorded at any session. The F1 average peaked at S2 (0.397 oct) and returned to its lowest value at S4 (0.200 oct), but

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this apparent convergence is driven almost entirely by [ɜː]; the remaining vowels showed no consistent trend. F2 distances showed no convergence toward the NH reference: the F2 average increased monotonically from S1 (0.138 oct) to S4 (0.200 oct), moving further from rather than closer to NH. No Spearman correlation reached conventional significance (all $p \geq .083$), with the exception of the F2 average ($\rho = +1.00$, $p = .083$). For most vowels, F1 and F2 Wasserstein distances tended to co-vary across sessions: when F1 displacement increased, F2 followed, and vice versa, suggesting global vowel shifts rather than formant-specific reorganisation. [ɜː] was the clear exception, with a highly vowel-specific F1 trajectory that was largely independent of its F2 trajectory. Throughout, Case Study 1's distances remained broadly comparable to the Post-LCI reference, though with substantially greater session-to-session variability, consistent with the noisier individual trajectory expected early in adaptation.

Formant Centroid Shift. Figure 5.5 shows the formant centroid shift trajectory and Table 5.4 reports numerical values.

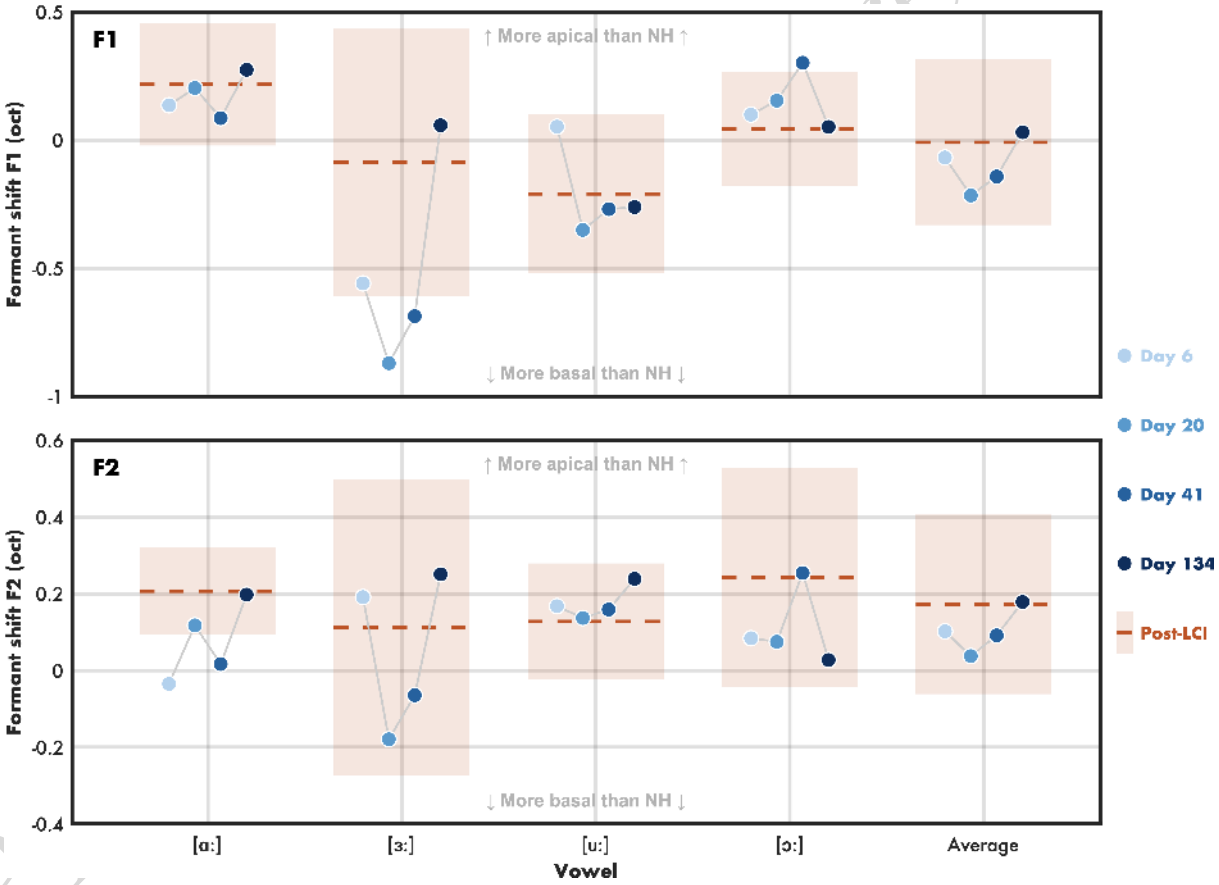


Figure 5.5: **Formant centroid shift relative to the NH reference across sessions (Case Study 1).** Shift $= \log_2(\hat{f}_{\text{NH}}/\hat{f}_{\text{session}})$ in octaves; positive values indicate the session centroid is perceived at a more apical place than for NH (lower frequency). Zero marks the NH reference; the red dashed line with shaded band shows the Post-LCI mean ± SD.

Table 5.4: **Formant centroid shift (oct), Case Study 1.** Positive = perceived at a more apical place than NH (lower frequency); negative = more basal (higher frequency) than NH.

Formant	Vowel	S1 (d6)	S2 (d20)	S3 (d41)	S4 (d134)
F1	[ɑː]	+0.136	+0.203	+0.086	+0.274
	[ɜː]	−0.559	−0.871	−0.687	+0.058
	[uː]	+0.052	−0.351	−0.270	−0.261
	[ɔː]	+0.099	+0.154	+0.302	+0.051
	Average	−0.068	−0.216	−0.142	+0.031
F2	[ɑː]	−0.035	+0.118	+0.017	+0.198
	[ɜː]	+0.191	−0.180	−0.065	+0.251
	[uː]	+0.168	+0.137	+0.159	+0.239
	[ɔː]	+0.084	+0.075	+0.254	+0.027
	Average	+0.102	+0.037	+0.091	+0.179

The shift analysis sharpens the co-movement pattern observed in the Wasserstein trajectories. For most vowels, F1 and F2 shifts evolved in the same direction across sessions: for example, for [ɜː] between S3 and S4, F1 shifted from −0.687 to +0.058 oct (moving from basal to slightly apical relative to NH) and F2 moved in the same direction from −0.065 to +0.251 oct, both shifting towards more apical representations simultaneously. For F2, values remained consistently positive across sessions (average: +0.102 at S1, +0.037 at S2, +0.091 at S3, +0.179 at S4), indicating that F2 representations were persistently perceived at a more apical place than for NH - that is, at a lower perceived frequency than the NH reference - with a slight increase over the observation window. For F1, the average trajectory moves from slightly negative (−0.068 oct at S1) through a trough at S2 (−0.216 oct) and back towards zero at S4 (+0.031 oct), superficially suggesting convergence; however, this trend is driven almost entirely by [ɜː], and given that F2 is simultaneously drifting away from NH there is no basis to conclude that F1 convergence will continue beyond Day 134. The Post-LCI reference band indicates that the experienced group's F2 centroid is consistently more apical than NH (positive values), consistent with the persistently positive F2 values observed in Case Study 1 from Day 6 - suggesting that the characteristic apical F2 shift is established from the very first session rather than developing progressively, and that both Case Study 1 and the Post-LCI group are displaced in the same direction relative to NH.

F1/F2 Reliance Ratio. Figure 5.6 shows the reliance ratio trajectory per vowel and session.

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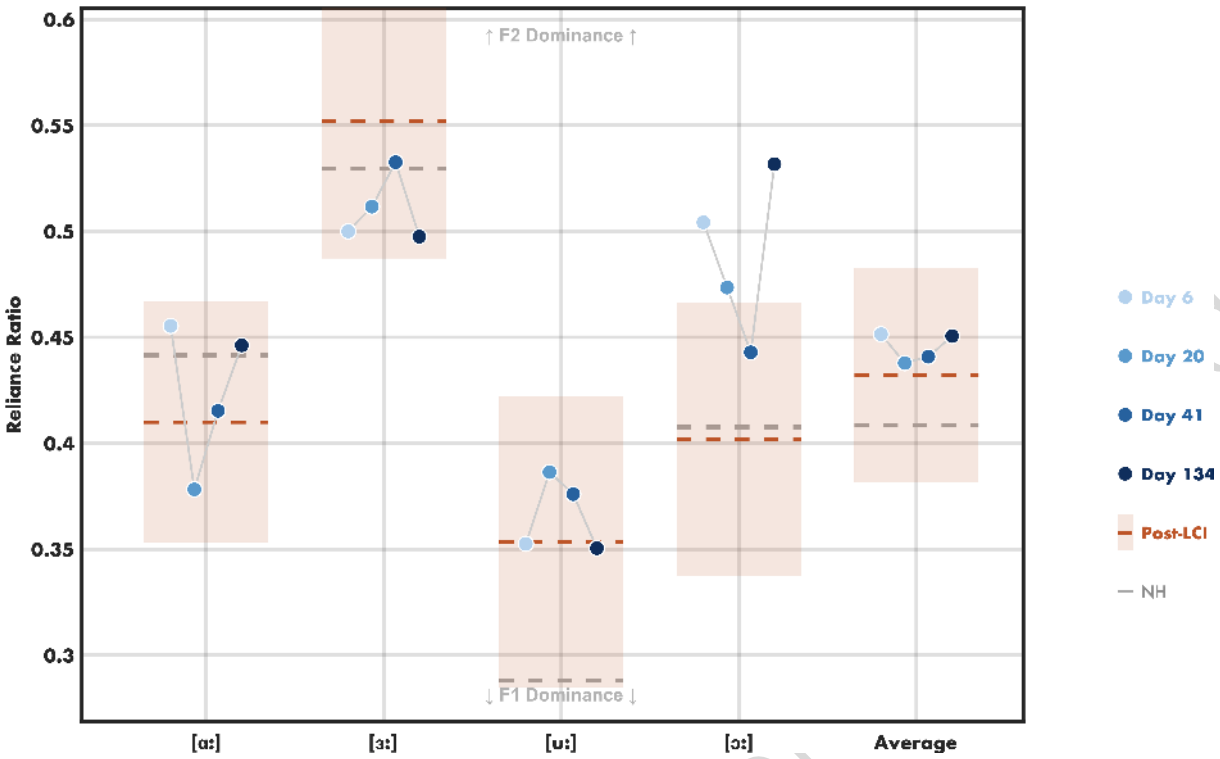


Figure 5.6: **F1/F2 reliance ratio per vowel and session (Case Study 1).** $R = \text{peak}(\text{bell}_{F2})/(\text{peak}(\text{bell}_{F1}) + \text{peak}(\text{bell}_{F2}))$. Values below 0.5 indicate F1 dominance. The grey dashed line shows the NH reference per vowel; the red dashed line with shaded band shows the Post-LCI mean ± SD.

No clear temporal trend was observed in the reliance ratio for any vowel. The mean ratio ranged narrowly from 0.438 to 0.451 across sessions (NH: 0.409), indicating mild and stable F1 dominance throughout. The profile at Day 6 was already comparable to that at Day 134, and no vowel showed a clear drift toward or away from the NH reference over the observation window. The Post-LCI mean reliance ratio was 0.432 (average across vowels), closely matching Case Study 1's values at every session and confirming that the balanced F1/F2 weighting strategy is established from Day 6 at a level consistent with the Post-LCI group. The relative weighting of F1 and F2 information was therefore established from the very first session and was not the primary locus of adaptation.

Question 2: Do Frequency-to-Place Factors Predict the Electrode-Level Adaptation Trajectory for the Vowel Task?

Delta confidence analysis. Figure 5.7 shows the delta confidence as a function of each factor per session, and Table 5.5 reports LMM results.

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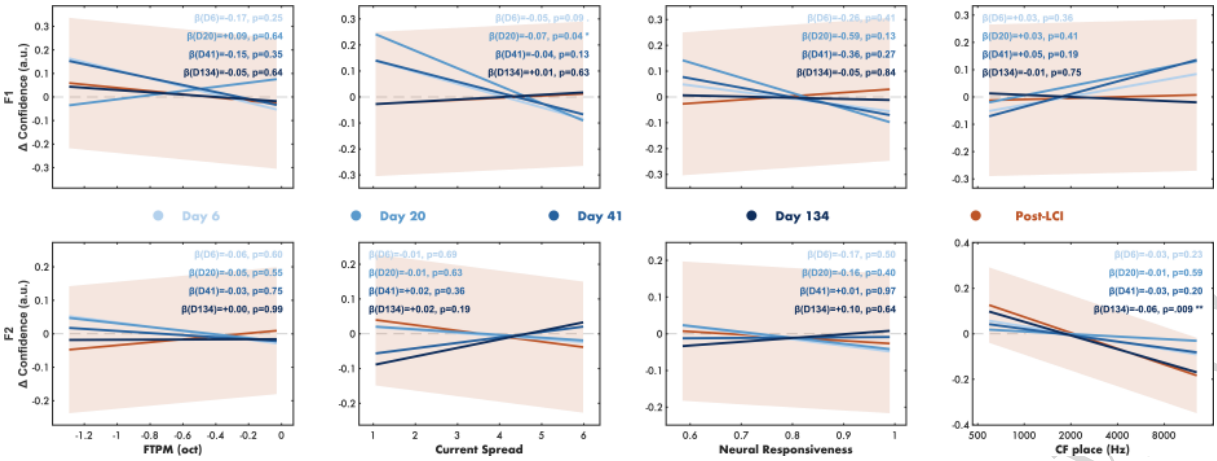


Figure 5.7: **Local confidence deficit (Δ = bell$_{\text{session}}$ − bell$_{\text{NH}}$) as a function of electrode-level factors across sessions (Case Study 1).** Each panel shows one factor (columns: FTPM = frequency-to-place mismatch in octaves; CS = current spread; NR = neural responsiveness; CF$_{\text{place}}$ = characteristic frequency of the stimulated cochlear location) and one formant (rows: F1, F2) across sessions (light to dark shading = Day 6 to Day 134). A value of zero indicates no deviation from the NH reference at that frequency location. The Post-LCI group mean trend and ±1 SD band are shown in red.

Table 5.5: **LMM $\hat{\beta}$ per session for each factor and formant (Case Study 1).** Model: $\Delta \sim$ Factor + Vowel + (1|Electrode). FTPM = frequency-to-place mismatch (oct); CS = current spread; NR = neural responsiveness; CF$_{\text{place}}$ = characteristic frequency of stimulated cochlear location. Significance: $^{\cdot}p < .10$, $^{*}p < .05$, $^{**}p < .01$.

Formant	Factor	Day 6	Day 20	Day 41	Day 134
F1	FTPM	−0.172	+0.088	−0.148	−0.049
	CS	−0.046$^{\cdot}$	−0.067*	−0.042	+0.009
	NR	−0.258	−0.593	−0.365	−0.045
	CF$_{\text{place}}$	+0.030	+0.034	+0.046	−0.007
F2	FTPM	−0.064	−0.055	−0.031	+0.002
	CS	−0.009	−0.008	+0.016	+0.025
	NR	−0.174	−0.162	+0.009	+0.103
	CF$_{\text{place}}$	−0.033	−0.011	−0.028	−0.060**

Only two effects reached significance across the full delta analysis: CS predicted F1 delta at Day 20 ($\hat{\beta} = -0.067$, $p = .037$) and CF$_{\text{place}}$ predicted F2 delta at Day 134 ($\hat{\beta} = -0.060$, $p = .008$). Despite this near-complete null in formal significance, the F2 delta trajectories showed a consistent and interpretable pattern across all three primary factors. For FTPM, electrodes with greater absolute mismatch were initially associated with lower F2 confidence relative to NH ($\hat{\beta} = -0.064$ at Day 6), but this deficit progressively attenuated and had fully resolved by Day 134 ($\hat{\beta} = +0.002$). For CS, electrodes with broader current spread were initially associated with lower F2 confidence ($\hat{\beta} = -0.009$ at Day 6), and this relationship reversed over time, with broader-current-spread electrodes attracting slightly more relative confidence by Day 134 ($\hat{\beta} = +0.025$), consistent with a progressive recalibration of spectral resolution expectations. For NR, electrodes

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with higher neural responsiveness were associated with lower F2 confidence at the earliest sessions ($\hat{\beta} = -0.174$ at Day 6), a pattern that reversed substantially by Day 134 ($\hat{\beta} = +0.103$), suggesting that the perceptual advantage of high-NR electrodes emerges only after initial central recalibration rather than being immediately available from activation.

All three factors showed monotonic Spearman trends for CS and NR ($\rho = +1.00$, $p = .083$ each), and the convergent pattern across three independent factors suggests a real trajectory of early F2 deficit followed by progressive resolution, even if individual session effects fall short of significance in this single-participant design. For F1, no coherent temporal pattern was observed: estimates alternated in sign with no consistent direction for any factor, consistent with no structured factor effects on F1 confidence beyond $\mathrm{CF}_{\mathrm{place}}$.

5.3.2 Chord Preference Task

Descriptive Overview of Responses

Figure 5.8 shows the frequency importance functions (FIFs) across the four sessions, and Figure 5.9 shows the distribution of preference grades per session.

At S1 (Day 6), the FIF was dominated by a strong mid-frequency preference: electrode E16 (999 Hz) was the most preferred (median = +3.93) and the four most basal electrodes (E1–E4, >4988 Hz) were the most strongly rejected (median = −3.52 to −1.71), yielding a total preference span of 7.45 units across the 22-electrode array. A broadly similar pattern was observed at S2 (Day 20), with E16 remaining the most preferred electrode (median = +2.85) and E1 the most rejected (median = −3.67), though the overall span had compressed to 6.52 units. By S3 (Day 41), the preference peak had shifted slightly apically to E17 (871 Hz, median = +2.64), and the most rejected electrode changed from the most basal (E1) to E4 (4988 Hz, median = −2.09), indicating a reorganisation of the electrode-level preference profile alongside a further reduction in span to 4.73 units. By S4 (Day 134), the FIF had become substantially less structured: the most preferred electrode was E11 (1933 Hz, median = +1.34), and the most rejected was E19 (621 Hz, median = −3.55) - an apical electrode that had been near-neutral at earlier sessions. The mean preference across electrodes declined progressively from +0.12 at S1 to −0.32 at S4, and the peak-to-trough span stabilised at approximately 4.9 units by S4, reflecting both a compression and a qualitative reorganisation of the preference profile rather than a simple flattening.

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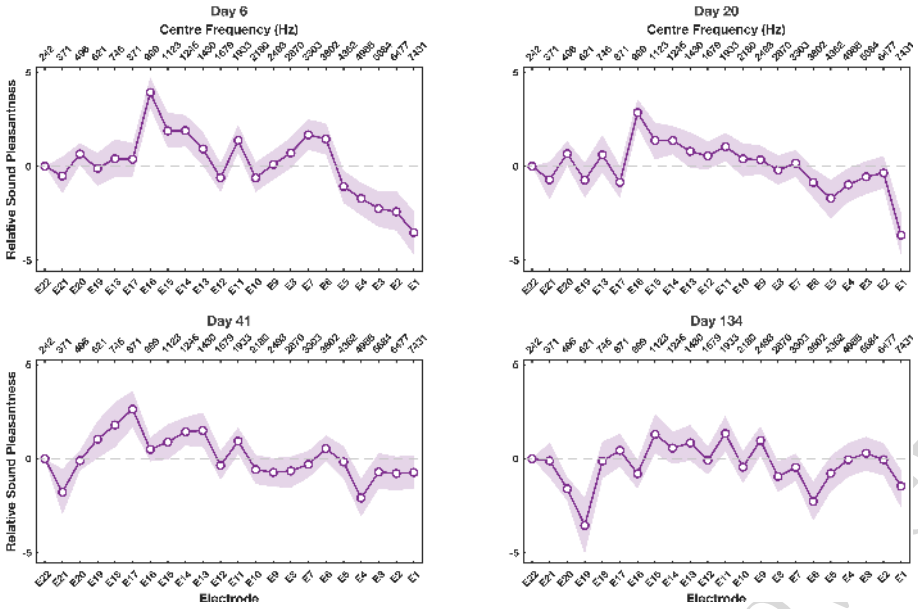


Figure 5.8: **Frequency importance function (FIF) across sessions (Case Study 1).** Each panel shows one testing session (Day 6, Day 20, Day 41, Day 134). The y-axis shows relative sound pleasantness (median with 16th–84th percentile shaded band); the bottom x-axis shows electrode labels (E = electrode number, E1 = most basal, E22 = most apical) and the top x-axis shows centre frequencies in Hz. The y-axis scale is shared across all four panels to allow direct comparison.

Grade distributions are centred around zero at all sessions, with a progressive compression of the range from S1 (grades spanning approximately ±3) to S4 (grades spanning approximately ±1), reflecting a reduction in the overall strength of preference with experience. The no-preference rate (grade = 0) accounted for 16, 21, 17, and 13 trials at S1–S4 respectively, remaining broadly stable whilst the non-zero grades became less extreme.

Root-position chords were preferred on 67.2%, 59.3%, 60.3%, and 53.7% of decided trials across sessions, and the less-distorted chord was preferred on 65.6%, 69.5%, 60.3%, and 56.7% of trials, both declining monotonically across sessions.

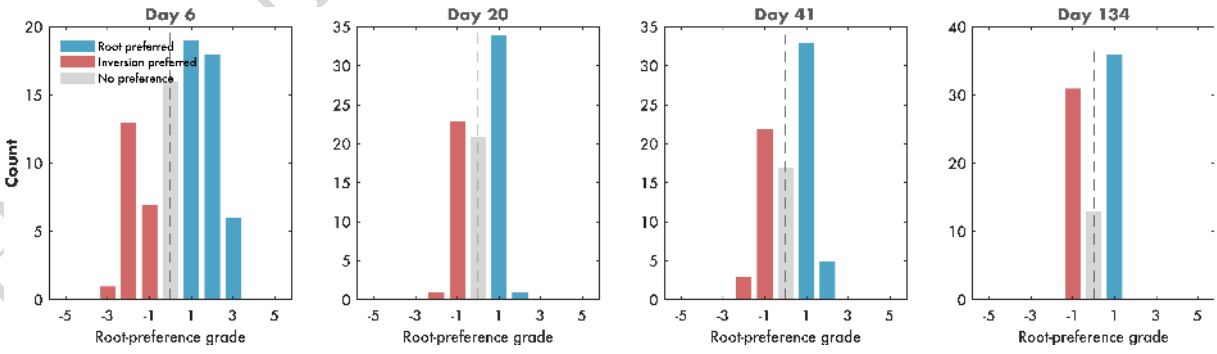


Figure 5.9: **Distribution of chord preference grades across sessions (Case Study 1).** Positive grades indicate preference for the root-position chord; negative grades indicate preference for the first-inversion chord. Grade = 0 indicates no preference. The progressive compression of the distribution from S1 to S4 reflects declining preference strength over the adaptation window.

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Question 3: Do Frequency-to-Place Factors Predict the Chord Preference Adaptation Trajectory?

Analysis 1 - Interval Distortion and Root-Position Preference

Per-Session Models. At S1, interval distortion (defined as the summed absolute difference between cochlear-place and allocated-frequency intervals; Equation 4.9) strongly predicted preference in the LMM (M-Add: $\hat{\beta}_{\Delta d} = -1.99$, $p = .003$) and root-position chords were significantly preferred (M-Base: $\hat{\beta}_{\text{root}} = +0.89$, $p = .012$), though the paired t-test on chosen versus rejected chord distortion was only marginal ($t(63) = -1.93$, $p = .058$). At S2, distortion sensitivity was maintained (M-Add: $\hat{\beta} = -1.23$, $p < .001$) and the paired t-test confirmed significantly lower distortion in chosen chords ($t(58) = -3.24$, $p = .002$), though root preference was no longer significant (M-Base: $p = .115$). At S3, neither distortion ($p = .306$) nor root preference ($p = .125$) reached significance in any model, and the paired t-test was null ($p = .662$). At S4, distortion sensitivity partially recovered (M-Add: $\hat{\beta} = -0.83$, $p = .027$) but root preference remained absent (M-Base: $p = .617$); instead, the FTPM gradient (the absolute inter-electrode FTPM difference) emerged as a significant predictor (M-Grad: $\hat{\beta}_{\Delta\text{FTPM}_{\text{gradient}}} = +1.92$, $p = .017$), with the paired t-test again marginal ($p = .060$). Figure 5.10 shows the per-session model coefficients and Figure 5.11 shows the corresponding chosen versus rejected chord distortion distributions.

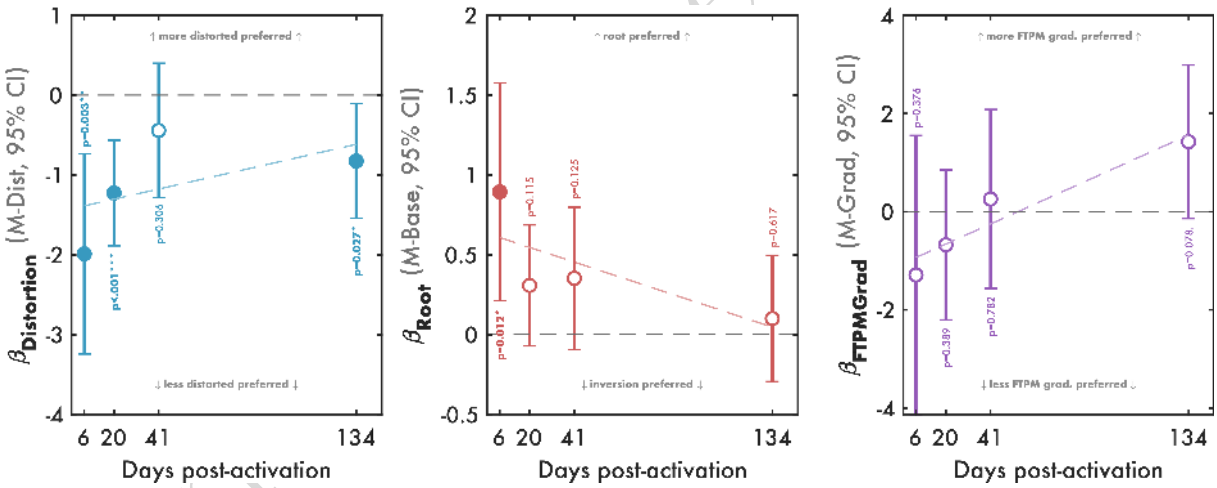


Figure 5.10: **Analysis 1: per-session regression coefficients for distortion sensitivity, root preference, and FTPM gradient (Case Study 1).** Distortion = summed absolute difference between cochlear-place and allocated-frequency intervals (Equation 4.9); FTPM gradient = absolute inter-electrode FTPM difference. Error bars show 95% confidence intervals. Coefficients are drawn from the best-fitting LMM at each session (M-Add at S1–S2, M-Base at S3, M-Grad at S4).

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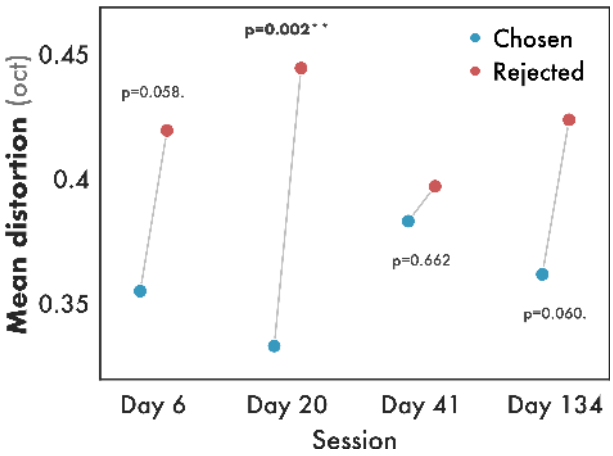


Figure 5.11: **Distortion of chosen versus rejected chords per session (Case Study 1).** Distortion = summed absolute difference between cochlear-place and allocated-frequency intervals. Each pair of dots shows the session mean distortion for chosen (teal) and rejected (red) chords, connected by a grey line. Chosen chords had systematically lower distortion than rejected chords at S1 ($p = .058$) and S2 ($p = .002$), but not at S3 ($p = .662$) or S4 ($p = .060$). Significance assessed by paired t-tests.

Temporal Change. The M-Add×Session LMM confirmed declining distortion sensitivity over time ($\hat{\beta}_{\text{Session}\times\Delta d} = +0.42$, $p = .040$) and the M-Base×Session LMM confirmed declining root preference ($\hat{\beta}_{\text{Session}\times\text{root}} = -0.23$, $p = .039$). FTPM gradient sensitivity increased significantly over time in the M-Grad×Session LMM ($\hat{\beta}_{\text{Session}\times\Delta\text{FTPM}_{\text{gradient}}} = +0.91$, $p = .049$). The combined M-Add×Session model confirmed both declining distortion sensitivity ($\hat{\beta} = +0.42$, $p = .039$) and declining root preference ($\hat{\beta} = -0.23$, $p = .034$) simultaneously.

Epoch Analysis. In the Early epoch (S1–S2), both distortion ($\hat{\beta} = -1.49$, $p = .001$) and root preference ($\hat{\beta} = +0.59$, $p = .002$) were significant in the M-Int LMM. In the Extended epoch (S3–S4), neither reached significance (distortion $p = .380$; root $p = .106$). In the Full epoch, both distortion ($\hat{\beta} = -0.93$, $p = .002$) and root preference ($\hat{\beta} = +0.41$, $p < .001$) were significant in the pooled M-Int model, though their interaction was not ($p = .372$). Figure 5.12 shows the trial-level distortion-preference relationship per session in the recoded frame, with the Post-LCI reference slope overlaid.

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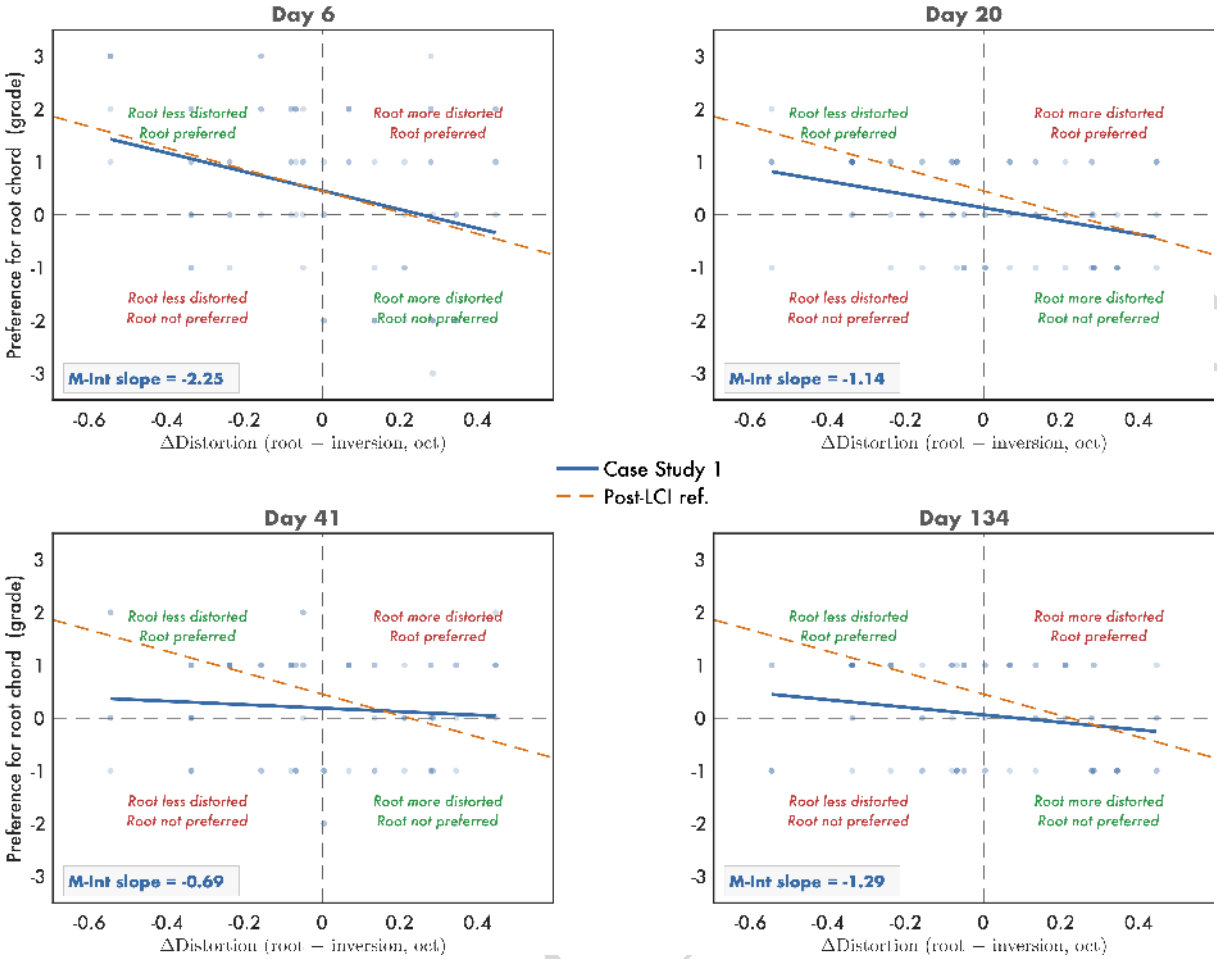


Figure 5.12: **Analysis 1: distortion-preference relationship per session (Case Study 1).** Each panel shows one testing session (Day 6, Day 20, Day 41, Day 134). All trials are recoded relative to the root-position chord: $x = \Delta d_{\text{root}} = d_{\text{root}} - d_{\text{inv}}$ (negative = root less distorted); y = preference for root chord (positive = root preferred). Distortion = summed absolute difference between cochlear-place and allocated-frequency intervals (Equation 4.9). The blue line is the per-session OLS fit; the dashed orange line shows the Post-LCI reference slope from Chapter 4 (M-Int simple slope, $\hat{\beta} = -2.03$, intercept = +0.45). The M-Int simple slope for each session is annotated bottom-left. Green quadrant labels indicate the two quadrants consistent with distortion avoidance; red labels indicate the opposite pattern.

Analysis 2 - Electrode Factor Prediction of Chord Preference

The analysis was conducted at the electrode level ($n = 88$: 22 electrodes × 4 sessions). The linear LMM was preferred for all three predictors in the Full epoch (AIC: NR 287.6, CS 288.4, FTPM 286.8) and is treated as the primary test. Figure 5.13 summarises the epoch-level interaction coefficients for all three predictors. In addition to the longitudinal interaction analysis, per-session cross-sectional OLS slopes are reported alongside the Post-LCI group reference slope to directly benchmark Case Study 1 against the experienced-user asymptote at each session.

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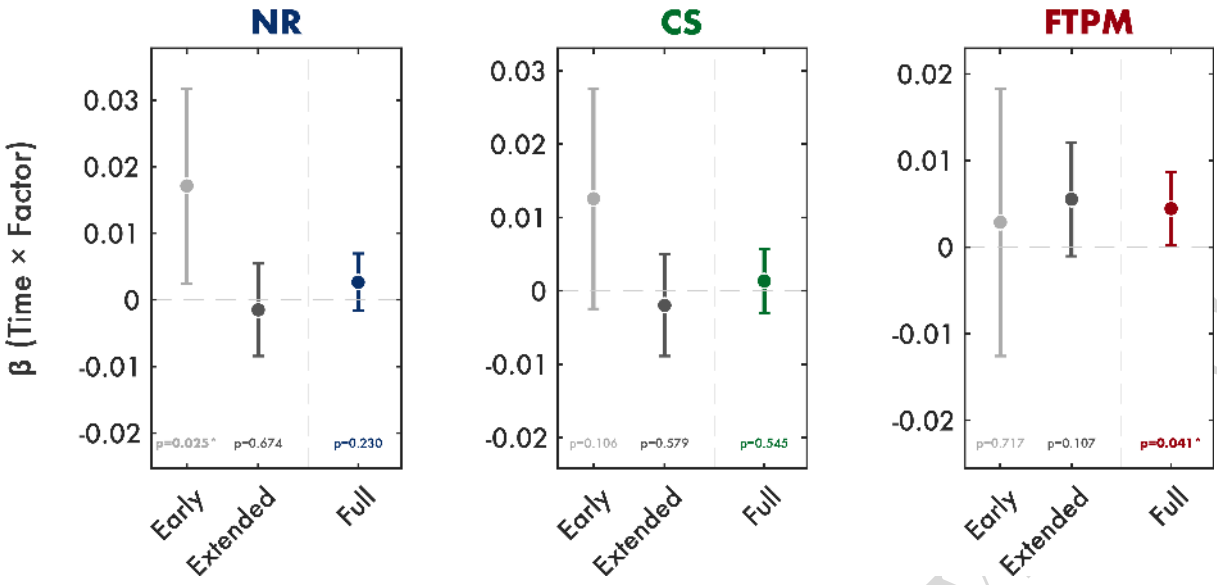


Figure 5.13: **Analysis 2: epoch-level forest plot of predictor×Time interaction coefficients (Case Study 1).** Each row shows the $\hat{\beta}$ (day^{-1}) and 95% CI for one predictor in one epoch (Early: Days 6–41; Extended: Days 41–134; Full: Days 6–134). NR = neural responsiveness; CS = current spread; FTPM = frequency-to-place mismatch. A positive coefficient indicates that preference for chords associated with higher values of that predictor increased over time within that epoch. Filled markers denote $p < .05$.

NR. In the Full epoch, the NR×Time interaction was not significant ($\hat{\beta} = +0.0027$/day, $SE = 0.0022$, $p = .230$). In the Early epoch (Days 6–41, $n = 66$), the interaction was significant and positive ($\hat{\beta} = +0.0171$/day, $SE = 0.0075$, $p = .025$): electrodes with higher neural responsiveness attracted progressively stronger preference from S1 to S3. In the Extended epoch (Days 41–134), the interaction reversed in sign and was non-significant ($\hat{\beta} = -0.0015$/day, $p = .674$), indicating that the early NR-driven preference increase did not persist into the later phase. The per-session OLS slopes reveal that the NR-preference relationship was initially strongly negative at S1 ($\hat{\beta} = -5.61$, $r = -.483$, $p = .023$), meaning that electrodes with higher NR were the least preferred at the earliest session. This relationship converged monotonically towards the Post-LCI reference ($\hat{\beta}_{\text{ref}} = +0.067$, $p = .959$) - which itself is indistinguishable from zero - across Sessions S2 ($\hat{\beta} = -4.20$) and S3 ($\hat{\beta} = -1.45$, $p = .421$), before diverging slightly at S4 ($\hat{\beta} = -2.43$, $p = .173$). The convergence towards the Post-LCI null is therefore the primary longitudinal signal: Case Study 1's early negative NR effect dissolves progressively over the first 41 days, consistent with the auditory system learning to exploit high-NR electrodes.

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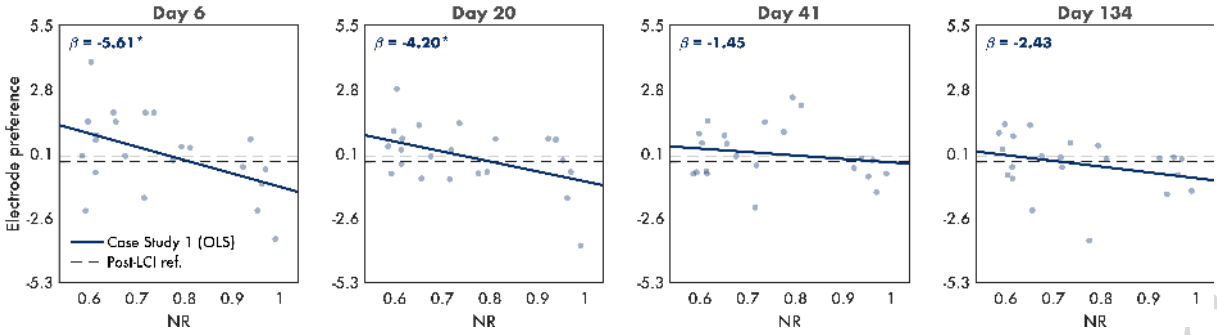


Figure 5.14: **Analysis 2: neural responsiveness (NR) and electrode preference across sessions (Case Study 1).** Each panel shows one testing session (Day 6, Day 20, Day 41, Day 134). Each dot represents one electrode (E = electrode); the solid blue line is the per-session OLS fit; the dashed black line shows the Post-LCI reference slope ($\hat{\beta}_{\text{ref}} = +0.067$, Chapter 4).

CS. CS showed no significant interaction in any epoch: Full ($\hat{\beta} = +0.0013$/day, $p = .545$), Early ($\hat{\beta} = +0.0126$/day, $p = .106$), and Extended ($\hat{\beta} = -0.0020$/day, $p = .579$). The per-session OLS slopes nonetheless show a coherent convergence pattern: the CS-preference slope at S1 ($\hat{\beta} = -0.411$, $p = .053$) exceeded the Post-LCI reference ($\hat{\beta}_{\text{ref}} = -0.212$, $p = .077$) in magnitude by a factor of approximately two, and converged steadily towards it across sessions, reaching $\hat{\beta} = -0.344$ at S2, $\hat{\beta} = -0.155$ at S3, and $\hat{\beta} = -0.263$ at S4 (delta from reference $= -0.051$). CS therefore shows the closest convergence to the Post-LCI asymptote of all three factors by Day 134, despite the absence of a significant longitudinal interaction, and the direction of the effect is consistent across all four sessions.

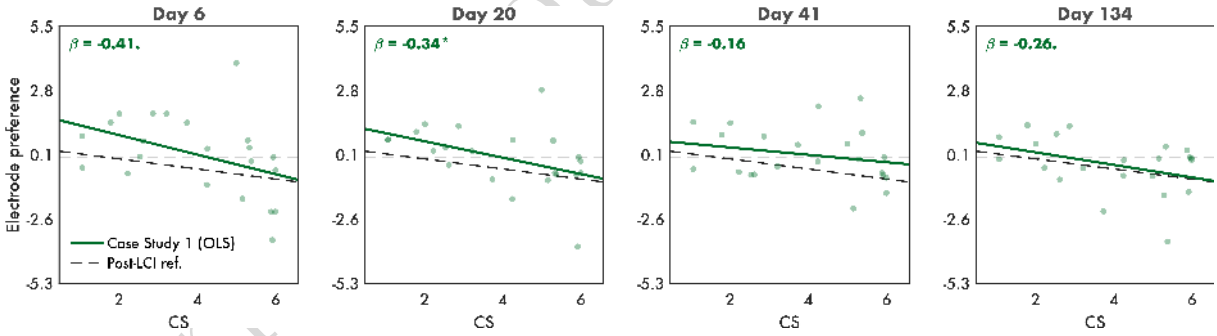


Figure 5.15: **Analysis 2: current spread (CS) and electrode preference across sessions (Case Study 1).** Same layout as Figure 5.14. The solid green line is the per-session OLS fit; the dashed black line shows the Post-LCI reference slope ($\hat{\beta}_{\text{ref}} = -0.212$, Chapter 4). No significant CS×Time interaction was observed in any epoch (all $p > .10$), yet the per-session slopes track the Post-LCI reference closely by Day 134.

FTPM. FTPM showed a significant positive interaction in the Full epoch ($\hat{\beta} = +0.0045$/day, $SE = 0.0022$, $p = .041$): electrodes with greater absolute FTPM were associated with progressively increasing chord preference across the full 134-day window. This effect was absent in the Early epoch ($\hat{\beta} = +0.0029$/day, $p = .717$) and marginal in the Extended epoch ($\hat{\beta} = +0.0055$/day, $p = .107$), indicating that the FTPM effect accumulated gradually. The per-session OLS slopes reveal a non-monotonic trajectory relative to the Post-LCI reference ($\hat{\beta}_{\text{ref}} = +0.835$, $p = .035$): Case Study 1 began well above the reference

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at S1 ($\hat{\beta} = +2.22$, $p = .061$, delta = +1.39), converged substantially by S2 ($\hat{\beta} = +1.35$, delta = +0.51), then diverged at S3 where FTPM was the only factor to reach significance ($\hat{\beta} = +1.80$, $p = .026$, delta = +0.96), before collapsing substantially below the reference at S4 ($\hat{\beta} = +0.11$, $p = .902$, delta = −0.73). This non-monotonic pattern - initially overshooting the Post-LCI asymptote and then undershooting it - is consistent with a system in which mismatch-driven electrode contributions are initially over-weighted, briefly re-emerge as the electrode-based representation consolidates at S3, and then recede as the preference profile loses its FTPM gradient by Day 134.

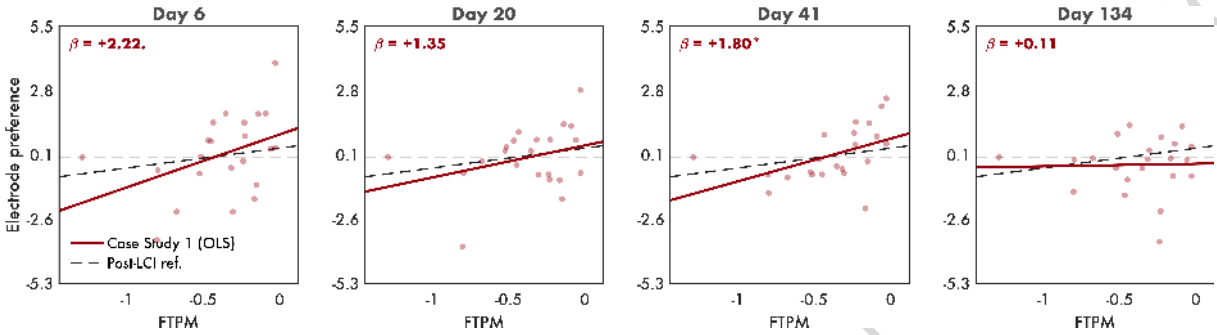


Figure 5.16: **Analysis 2: frequency-to-place mismatch (FTPM) and electrode preference across sessions (Case Study 1).** Same layout as Figure 5.14. The solid red line is the per-session OLS fit; the dashed black line shows the Post-LCI reference slope ($\hat{\beta}_{\text{ref}} = +0.835$, Chapter 4). A significant positive FTPM×Time interaction in the Full epoch ($\hat{\beta} = +0.0045$/day, $p = .041$) reflects a non-monotonic trajectory: Case Study 1 overshoots the Post-LCI reference at S1 and S3, converges at S2, and drops below the reference at S4.

5.4 Discussion

This chapter tracked the progression of the perceptual representations of a newly implanted post-lingual adult CI user across four sessions spanning 6 to 134 days post-activation, using the vowel identification task from Chapter 3 and the chord preference paradigm from Chapter 4. The two tasks were administered at the same sessions, enabling a direct comparison of how the adaptation process unfolds across two fundamentally different perceptual domains. The vowel and chord findings are discussed separately before being brought together in a comparative interpretation.

5.4.1 Vowel Task

Question 1: What does the Adaptation Trajectory Look Like for the Vowel Task?

Conjoint F1/F2 Movement and Formant Displacement A striking feature of the Wasserstein and shift trajectories is that F1 and F2 distances tend to co-vary across sessions for most vowels. For [ɑː], [uː], and [ɔː], increases in F1 displacement were accompanied by increases in F2 displacement, and decreases similarly paired, with both formants drifting in the same direction across sessions. This co-movement is inconsistent with a

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formant-specific learning account, in which adaptation would be expected to affect the F1 and F2 regions independently. An alternative interpretation is that the co-movement is driven by one formant specifically - with the other following - rather than reflecting genuinely global recalibration. This possibility is partly supported by the factor analyses reported in Question 2: if electrode-level factors at a specific formant region are guiding the temporal trajectory, the co-movement of the other formant may be a consequence rather than a primary adaptation signal. What can be said from the present data is that the two formants do not adapt independently, and that the primary centre of change appears to be F2 given its structured relationship with individual factors across sessions.

Notably, the reliance ratio showed no temporal trend across any vowel, with the Day 6 profile already indistinguishable from Day 134. Case Study 1 arrived at implantation with a balanced F1/F2 weighting strategy - F1 slightly dominant on average, consistent with Post-LCI users in Chapter 3 - suggesting that spectral weighting is pre-existing and is not a mechanism of early adaptation.

F2 is Persistently Perceived at a More Apical Place than for NH The F2 centroid shift was consistently positive across all sessions (average +0.102 to +0.179 oct), indicating that F2 representations were persistently perceived at a more apical place than for NH throughout the observation window. Importantly, the Post-LCI reference band shows that the experienced group F2 centroid is also more apical than NH, suggesting that Case Study 1's apical F2 placement is not a transient early feature but is present at Day 6. The F2 trajectory therefore describes progressive consolidation of an apical bias already present at activation rather than a drift away from NH. This is broadly consistent with the view that post-lingual CI users adapt their perceptual frequency map towards the actual cochlear stimulation sites over the first months of device use, with the direction and magnitude of the shift depending on the individual electrode placement and frequency allocation.

The F1 average trajectory (from −0.068 oct at S1 through a trough of −0.216 oct at S2 back towards zero at S4) superficially suggests convergence, but this is driven almost entirely by [ɜː]. Its large and non-monotonic F1 trajectory - reaching −0.871 oct at Day 20 before recovering to near zero at Day 134, whilst its F2 trajectory remained comparatively flat - is the signature of genuine electrode-to-category learning for a single vowel category, independent of the global session effect. The fact that [ɜː] - which has the highest F1 frequency of the four retained vowels - was the vowel undergoing this correction is consistent with mid-array electrodes being the most dynamically reorganised during early adaptation. Excluding [ɜː], there is no convergence trend for F1, and whether F1 would continue converging beyond Day 134 remains unknown.

Question 2: Do Frequency-to-Place Factors Predict the Electrode-Level Adaptation Trajectory for the Vowel Task?

Delta Confidence: Early F2 Deficits and Progressive Resolution The delta confidence analysis reveals a structured temporal pattern for F2 despite failing to reach statistical significance, and constitutes the most informative finding of the vowel experiment. Only two effects reached significance across the full analysis: CS predicted F1 delta at Day 20 ($\hat{\beta} = -0.067$, $p = .037$) and $\mathrm{CF}_{\mathrm{place}}$ predicted F2 delta at Day 134 ($\hat{\beta} = -0.060$,

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$p = .008$). Despite this near-complete null, a convergent pattern emerged across all three primary F2 factors.

For FTPM, electrodes with greater absolute mismatch were initially associated with lower F2 confidence relative to NH ($\hat{\beta} = -0.064$ at Day 6), but this deficit progressively attenuated and had fully resolved by Day 134 ($\hat{\beta} = +0.002$): the participant placed progressively more confidence towards electrodes with less absolute mismatch over time, consistent with a refinement of spectral coding around lower-FTPM regions as the frequency map is progressively recalibrated (Fu and Galvin, 2007b). For CS, electrodes with broader current spread were initially associated with lower F2 confidence ($\hat{\beta} = -0.009$ at Day 6), and this relationship reversed over time ($\hat{\beta} = +0.025$ at Day 134): rather than avoiding broader-current-spread electrodes, the listener progressively learned to extract useful information from them, suggesting a dynamic rather than static consequence of CS on perception that is not captured in cross-sectional analyses of experienced users. For NR, the pattern is also striking: electrodes with higher neural responsiveness were associated with lower F2 confidence at the earliest sessions ($\hat{\beta} = -0.174$ at Day 6), which reversed substantially by Day 134 ($\hat{\beta} = +0.103$). NR, as estimated by PECAP, reflects the relative synchronous responsiveness of the auditory nerve at each electrode compared with other array positions (Garcia et al., 2021): its paradoxical early negative relationship with confidence may reflect the fact that high-NR electrodes produce more salient but initially unfamiliar percepts, and that their perceptual advantage only becomes exploitable after central recalibration. This is consistent with broader evidence that the quality of the electrode-neuron interface is a strong predictor of speech perception outcomes but that this advantage takes time to be expressed behaviourally (He et al., 2024; Pfingst et al., 2015). Monotonic Spearman trends were observed for both CS and NR ($\rho = +1.00$, $p = .083$ each).

This convergent pattern across three independent factors is the strongest longitudinal signal in the vowel data, and suggests that it is F2 - rather than F1 - that is primarily structured by electrode-level factors across sessions. Whether this finding generalises across participants, and whether the reliance ratio proves sensitive to such reorganisation over a longer adaptation window, will need to be examined in the ongoing longitudinal cohort.

5.4.2 Chord Task

Question 3: Do Frequency-to-Place Factors Predict the Chord Preference Adaptation Trajectory?

Early Sensitivity: Pre-Existing Acoustic Templates Drive Preference At S1 and S2 (Days 6 and 20), Case Study 1 showed clear sensitivity to interval distortion and strong root-position preference, both consistent with the application of pre-existing acoustic spectral templates built through prior NH experience (Reiss et al., 2007). Chords whose physical interval structure most closely matched the intended acoustic intervals were preferred, and root-position chords were preferred over inversions. These effects declined rapidly: by S3 (Day 41), neither distortion nor root preference reached significance in any model.

The disappearance of these effects by 41 days does not necessarily reflect improved

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adaptation in the conventional sense. A more plausible account is that the acoustic spectral templates driving early sensitivity were progressively suppressed as the central auditory system began to represent chord stimuli in terms of the actual electrode positions being stimulated rather than the acoustic frequencies they nominally represented. Once chord identity is encoded primarily at the electrode level, interval distortion defined in acoustic frequency space becomes less perceptually relevant. This transition from acoustic to electrode-based coding is consistent with the broader literature on perceptual adaptation to FTPM: post-lingual CI users who begin with strong acoustic templates must progressively recalibrate their perceptual frequency map before the electrode-level representation can organise perception (Fu and Galvin, 2007b; Svirsky et al., 2010). The six-week transition point identified here is at the early end of the 3–6 month window typically reported for speech adaptation, which is consistent with the chord paradigm being more sensitive to early changes than standard speech tests. This is likely related to the special nature of speech perception: the cortical mechanisms underlying phoneme categorisation involve complex, language-specific boundary learning and dynamic adaptation to different speakers, making the speech perception system inherently resilient to small spectral changes (Hillenbrand et al., 1995; Mathew et al., 2018). This resilience means that speech-based measures are likely insensitive to the kind of electrode-level spectral reorganisation detected here by the chord paradigm; however, when speech-based differences are detected, they are likely to be perceptually important precisely because they have exceeded this high tolerance threshold. An alternative interpretation is that the increased ability to perceptually distinguish between electrodes - rather than adaptation per se - underlies the loss of template-driven preference: as electrode-level representations become more distinct, global acoustic-based judgements give way to more fine-grained local comparisons.

Late Emergence: Electrode-Level FTPM Takes Over By S4 (Day 134), the FTPM gradient (the absolute difference in FTPM between the two chord electrodes) emerged as the primary predictor of chord preference ($\hat{\beta} = +1.92$, $p = .017$), indicating a shift in the perceptual basis of preference: the listener was no longer evaluating chords against an acoustic template but against an electrode-level representation in which apically displaced electrodes contributed more salient percepts. This is consistent with recent evidence that FTPM-induced interval distortion is the primary mechanism by which music perception is degraded in CI users, and that this distortion operates specifically through the compression of pitch intervals at the cochlear stimulation sites (Lewis et al., 2025).

The electrode-level trajectory analysis confirmed this two-phase picture. NR (neural responsiveness) showed a significant positive interaction in the Early epoch (S1–S3, $\hat{\beta} = +0.017$/day, $p = .025$): electrodes with better neural health attracted progressively stronger preference in the first six weeks, consistent with NR being associated with the formation of reliable chord percepts as the system transitions from acoustic to electrode-based representation (Garcia et al., 2021; Pfingst et al., 2015). FTPM (frequency-to-place mismatch) showed a significant positive interaction in the Full epoch ($\hat{\beta} = +0.0045$/day, $p = .041$), with no significant early effect, confirming that the FTPM effect accumulated gradually and only became detectable across the full 134-day window. CS (current spread) showed no significant effect at any stage.

The dissociation between NR (Early) and FTPM (Late) is theoretically informative. In the early weeks, the most reliable perceptual representations are formed at electrodes

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with the best neural health, providing the auditory system a stable baseline around which to organise chord percepts (He et al., 2024). As electrode-based representation consolidates in the Extended phase, FTPM becomes the dominant organisational factor, with electrodes with greater absolute FTPM gaining the most preference improvement over time.

When compared against the cross-sectional Post-LCI group, the per-session slopes suggest a global trend of convergence towards the experienced-user asymptote, though the three factors follow distinct trajectories. For NR, Case Study 1 began with a strongly negative per-session slope ($\hat{\beta} = -5.61$ at S1), meaning high-NR electrodes were the least preferred; this converged progressively towards the Post-LCI reference of approximately zero ($\hat{\beta}_{\text{ref}} = +0.067$, $p = .959$) across S2 and S3, before diverging slightly at S4 - consistent with the sign reversal observed in the delta analysis. For CS, convergence was the most systematic: the per-session slope moved steadily from -0.411 at S1 towards the Post-LCI reference of -0.212, reaching -0.263 at S4 (delta $= -0.051$), the closest alignment of any factor. For FTPM, the trajectory was non-monotonic: Case Study 1 initially overshot the Post-LCI reference ($\hat{\beta} = +2.22$ vs $\hat{\beta}_{\text{ref}} = +0.835$ at S1), converged at S2, diverged at S3 when FTPM was the only factor to reach significance ($\hat{\beta} = +1.80$, $p = .026$), then dropped substantially below the reference at S4 ($\hat{\beta} = +0.11$). This non-monotonic pattern is reminiscent of the incomplete adaptation described in modelling studies of severely shifted frequency maps (Svirsky et al., 2010), and suggests that the FTPM effect does not follow a simple monotonic trajectory but instead reflects a system that transiently amplifies mismatch-driven contributions before gradually stabilising. Taken together, these comparisons suggest that the experienced-user endpoint is not approached uniformly across factors or across sessions, but through a structured sequence in which NR effects resolve first, CS converges steadily, and FTPM follows a more complex trajectory that reflects its dual role as both a spatial organising principle and a source of perceptual instability during early adaptation.

5.4.3 Cross-Task Comparison

Placing the vowel and chord results alongside each other reveals both convergences and divergences that are theoretically informative.

Convergence: a Two-Phase Temporal Structure. Both tasks show evidence of a transition between an early phase governed by prior acoustic experience and a later phase governed by electrode-level properties. In the vowel task, F2 delta deficits are present early and attenuate progressively; in the chord task, acoustic template sensitivity is present early and gives way to electrode-level FTPM preference. In both cases, the transition occurs between S2 and S3 (Days 20 and 41), suggesting that approximately six weeks post-activation marks a shift in the perceptual representation across both speech and music domains. This timing is consistent with the early adaptation window identified in the broader literature on Post-LCI users, where perceptual learning is most rapid in the first months and eventually reaches an asymptote (Fu and Galvin, 2007b).

Convergence: NR is associated with early adaptation, FTPM organises late. For the prediction of the vowel adaptation trajectory based on electrode-level factors

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(FTPM, CS, NR), the delta analysis showed that all three factors imposed early F2 deficits at Day 6 that progressively attenuated, whilst the Post-LCI group showed NR as the sole residual predictor in the cross-sectional delta null. For the prediction of the chord adaptation trajectory based on the same electrode-level factors, the NR-Early and FTPM-Late pattern was more clearly delineated: NR was associated with the formation of reliable chord percepts in the first six weeks, and as the auditory system integrated more electrodes into its representation, FTPM - as a spatial mapping property - became the dominant organising principle. This parallel across tasks suggests a common underlying mechanism: NR provides the quality of the initial neural baseline (Garcia et al., 2021), and FTPM progressively structures the electrode-level representation once that baseline is established.

Divergence: the Direction of the FTPM Effect. At Day 134, electrodes with greater absolute FTPM simultaneously attracted higher chord preference (Analysis 2), yet in the delta analysis the same electrodes showed lower F2 confidence relative to NH. These two findings from the same electrodes at the same session can be reconciled by what each measure captures: chord preference is a relative evaluation rewarding perceptual intensity, whilst delta reflects local deficit relative to NH. Apical electrodes stimulate cochlear regions the auditory system finds reliable in absolute terms and perceptually vivid in relative terms, but because NH listeners would place even more confidence at those cochlear positions, the delta remains negative. This dissociation describes tolerance of mismatch rather than correction: the listener learns to use mismatched apical electrodes productively without correcting their frequency-to-place error (Svirsky et al., 2010).

Divergence: Task Sensitivity to Adaptation. The chord task showed clear and statistically significant changes across sessions, whilst the vowel task produced only sub-threshold trends. This difference in sensitivity is likely intrinsic to the paradigm designs rather than to the underlying perceptual process: the paired-comparison format of the chord task is inherently more sensitive than the closed-set identification format of the vowel task, and the chord outcome variable - preference grade - is a direct relative judgement that does not require formant estimation, Bayesian posterior computation, or confidence weighting. This is consistent with the superior sensitivity of the chord paradigm observed in the Post-LCI group from Chapter 4, where distortion effects that were marginal or absent in pooled models became detectable once the interaction with chord type was modelled. The impaired perception of harmony and chord quality in CI users is well established (Zimmer et al., 2019; Gilbert et al., 2025), but the present longitudinal data are the first to show that this impairment is dynamically structured by electrode-level factors across time. These observations suggest that the chord paradigm is the more appropriate tool for tracking adaptation in real time, whilst the vowel bell framework provides a richer mechanistic characterisation of what has changed at the electrode level.

Limitations. The single-participant design is the primary limitation of this chapter. All findings are presented as theoretically motivated observations requiring replication in a larger longitudinal cohort with denser session sampling in the first six weeks. The formant shift metric is likely insensitive at early sessions when posterior confidence is near zero, and the delta null at Day 6 should be interpreted cautiously for this reason.

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5.5 Conclusion

This study tracked the perceptual representations of a newly implanted post-lingual CI user across four sessions spanning 6 to 134 days post-activation, using the vowel and chord paradigms developed in Chapters 3 and 4. Three main conclusions emerge.

Regarding how the vowel adaptation trajectory , F2 representations were persistently more apical than NH from Day 6 and showed no convergence across sessions, consistent with the asymptotic state already observed in Post-LCIs rather than a progressive trajectory towards it. F1 convergence was driven entirely by [ɜː], reflecting vowel-specific categorical learning rather than a global adaptation process. The F1/F2 reliance ratio was stable throughout, confirming that spectral weighting is a pre-existing property that does not require recalibration.

Regarding whether electrode-level factors predict the vowel adaptation trajectory, all three factors imposed early local F2 deficits relative to NH that attenuated progressively across sessions, contrasting with the complete null observed in Post-LCIs from Chapter 3. Case Study 1 was therefore at an intermediate stage of the adaptation resolution process at Day 134, with the deficit-attenuation pattern providing a longitudinal precursor to the experienced-user endpoint.

Regarding whether electrode-level factors predict the chord preference adaptation trajectory, the task revealed a clear two-phase structure. In the early phase (Days 6-20), preference was driven by acoustic spectral templates inherited from prior normal-hearing experience, with per-session factor slopes markedly exceeding the Post-LCIs reference. By Day 41 these effects had alleviated and by Day 134 all three factor slopes had converged towards the Post-LCIs asymptote, with NR scaffolding initial reliable percepts in the early phase and FTPM becoming the dominant spatial organising principle as electrode-based representation consolidated. These findings describe a process of tolerance rather than correction: the listener learns to use mismatched electrodes productively through a structured two-phase process that is substantially complete within the first six weeks of device use.

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Postface

The case study presented here is the first of a planned series of four newly implanted post-lingual adult CI users, for whom longitudinal data collection is ongoing. The findings reported above should therefore be read as hypothesis-generating rather than confirmatory: the two-phase temporal structure, the NR-Early and FTPM-Late dissociation, and the convergent F2 delta attenuation are theoretically motivated patterns that require replication across participants before inferential conclusions can be drawn.

What further participants will enable. With four participants, the session-level data can be pooled within a mixed-effects framework with participant as a random intercept, transforming the descriptive trajectories reported here into inferential estimates with proper uncertainty quantification. Crucially, between-participant variation in FTPM magnitude and pattern will allow a direct test of whether individuals with greater cochlear mismatch show slower or qualitatively different adaptation trajectories - a prediction that the frequency-to-place framework developed in this thesis motivates directly but that a single case cannot test. Whether the six-week transition point identified here (between S2 and S3) is a reliable marker of the acoustic-to-electrode representational shift, or an individual feature of Case Study 1, will only become clear across the larger cohort.

Limitations of the current paradigms. Both tasks carry limitations that the larger longitudinal study should address. For the vowel task, the use of a single signal at each trial and the conjoint estimation of F1 and F2 still limit the interpretation at the electrode level. Additionally, the reliance ratio remained stable from Day 6, suggesting that F1/F2 spectral weighting is established from activation in post-lingual users and is not a sensitive marker of early adaptation. This absence of temporal trend may partly reflect the way the ratio is calculated here, and alternative formulations should be considered in future analyses. For the chord task, structural differences between the root and the inverted chord presented at every trial also limit the interpretation. Other paradigms without these intrinsic acoustical constraints should be explored in the future.

Clinical implications and future directions. A key strength of the paradigms developed here is that they can be administered and interpreted at the individual level, yielding electrode-level profiles that are directly clinically actionable. For Case Study 1, the chord preference data at Day 134 suggest that the most basal electrodes (E1–E4) are consistently the least preferred, that the mid-frequency region (around E11–E16) carries the most perceptual weight, and that FTPM-driven preference has largely resolved by 4.5 months post-activation. On the basis of these data, no remapping intervention would be recommended for this participant at the current stage: the preference profile has stabilised, distortion sensitivity has attenuated, and the electrode-level factor associations are converging towards the experienced-user asymptote. However, had the chord task revealed persistent distortion sensitivity at Day 134 - particularly at specific electrode pairs with large FTPM gradients - targeted frequency reallocation at those pairs would have been a reasonable clinical recommendation.

More broadly, if the two-phase structure replicates across the cohort, it has a direct clinical implication: the acoustic template phase (Days 6–20) represents the window dur-

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ing which distortion avoidance and root-position preference are most salient, and during which targeted remapping - for example, reducing interval distortion at specific electrode pairs - would be most likely to interact with the listener's existing perceptual framework. By Day 41, the listener has begun to encode chords in electrode-level terms, and map adjustments at this stage may be less effective at altering the emerging electrode-based preference structure.

The chord paradigm, which proved more temporally sensitive than the vowel task, is a practical candidate for serial clinical assessment: it requires no metalinguistic training, produces reliable preference data within a single session, and yields electrode-level preference maps that can be compared directly to the individual factor profiles estimated from PECAP and post-operative imaging. Integrating these measures into a clinically deployable monitoring tool - in which adaptation trajectories are tracked against normative reference ranges derived from the full cohort - represents the primary translational goal of the larger longitudinal study.

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